

Homework #3

CSED501

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Due: **November 18, 2025**Name: Aeden Jameson

Instructions: Complete the questions below and upload a PDF containing your written answers to Gradescope.

- Keep your answers within the provided space.
- Answers can be handwritten, or typed and pasted into this PDF.
- Show your work; this makes giving partial credit easier!
- True/False questions can be answered with **True/False**, but explanations can be added if you feel they are needed.
- Numeric answers can be given in fraction or decimal form; if decimal, round to 2 places

Assigned reading (links can also be found in Canvas Week 4 Resources):

- Garg, Pratyush, John Villasenor, and Virginia Foggo. "Fairness metrics: A comparative analysis." 2020 IEEE international conference on big data (Big Data). IEEE, 2020.

Problem 1 7 pts: True/False. Mark each statement as True or False and briefly justify your answer.

- (a) PCA is effective when the data's underlying low-dimensional structure is non-linear.
- (b) If we are performing clustering, we typically assume we either do not have or do not use labels in training the model.
- (c) K-means centroids are always actual data points.
- (d) Initializing centroids using k-means++ guarantees convergence to a global optimum.
- (e) If the total value of the Within-Cluster Sum of Squares increases between two iterations of k-means, there is a bug in your k-means code.
- (f) Given a set of points in a d -dimensional space, using PCA to reduce the dataset to $d' < d$ dimensions will always lead to loss of explained variance.
- (g) GMMs assume that the data points within each component follow a Gaussian distribution.

Answer:

- a) False. PCA is designed to represent an original set of D features with a subset of k features where $k < D$. Those k features are a linear combination of the original D features with minimal reconstruction error. PCA components of a non-linear data set would lose more information and have a higher reconstruction error. Thus be less effective.
- b) True. Clustering is typically used to explore the structure of the data.
- c) False. Centroids are the mean of the points in the clusters, which may or may not be an actual data point.
- d) False. K++ just improves the initialization of the of the starting centroids.
- e) True. With each iteration the current point to assign will be assigned to the closest centroid. And then the mean, the new centroid, is re-computed for each cluster which minimize the WCSS.
- f) False. Not necessarily. You could have a trivial data set: $(1,0), (2,0), \dots, (n, 0)$. Applying PCA would have one component, x , that explains 100% of the variance.
- g) True. Yes, by definition.

Problem 2 4 pts: Consider the following matrix $X = \begin{bmatrix} 4 & 1 \\ 5 & 4 \\ 1 & 2 \\ 1 & 0 \end{bmatrix}$ with four data points in $\mathbb{R}^2$. We would like to use PCA to find a rank-1 linear representation of these data.

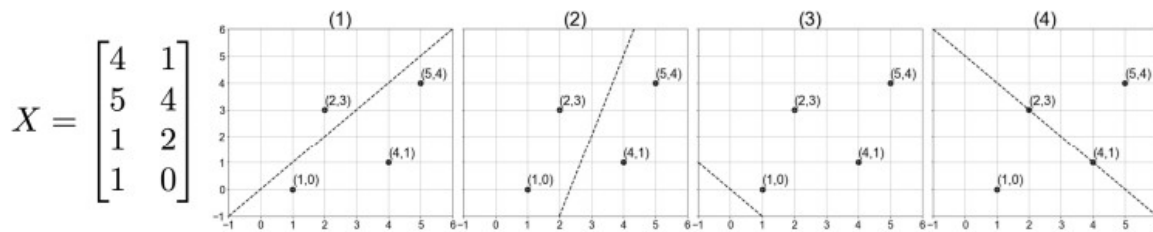


Figure 2: These plots depict data points from the four-sample dataset X .

Figure 1: 4 possible PCA directions.

Which line in Figure 1 represents the slope of the first principal component? Justify your answer.

Answer:

Assuming point x3 should actually be (2,3) then plot 1 represents the slope of the first principal component.

The actual first principal component passes through the sample mean which is

$$\text{mean} = ((4+5+2+1)/4, (1+4+3+0)/4) = (3, 2)$$

In addition we can observe from the sample data points that as x increases so does y therefore the candidate line must have positive slope. Now the general steepness can be assessed by joining the line of the two most distant points which leads to $(4 - 0) / (5 - 1) = 1$.

Using the facts above we can eliminate 3 & 4 because the lines have negative slope. We eliminate plot 2 because it's far too steep. Thus that leaves us with plot 1 whose line represents the slope of the principal.

Problem 3 **6 pts**: Which of the following facts could lead to misleading statistics in criminal justice datasets? Choose one and justify your answer.

- (a) Different reported crime rates across different neighborhoods.
- (b) Different wrongful arrest rates across different demographic groups.
- (c) Differences in local laws (e.g., different laws regarding indoor and outdoor drug sales in Seattle).
- (d) All of the above.

Answer:

All of the above.

- a) This could be due to differences in how much neighborhoods are policed.
- b) This can happen when the same law is more heavily enforced with specific groups.
- c) The example states why differences could lead to misleading stats.

Problem 4 12 pts: Consider the following dataset in $\mathbb{R}^2$:

Point	$x_i[1]$	$x_i[2]$
x_1	1	1
x_2	2	1
x_3	4	3
x_4	6	5
x_5	7	6
x_6	8	5

Start the centroid for cluster 1 at x_2 and the centroid for cluster 2 at x_6 . Perform the k-means algorithm with $k = 2$ until the cluster assignments do not change between successive iterations (convergence). Show your work.

- (a) What are the final cluster assignments for the data points? (Example notation for assignments: $\begin{bmatrix} 1 & 1 & 2 & 1 & 1 & 1 \end{bmatrix}$.)
- (b) How many iterations did the k-means algorithm need to converge? The initialization does not count as an iteration.

Answer:

a) $x_n \rightarrow c_1$ or $c_2 = Y$ represents the squared euclidean distance between those points. There's not enough room for all the math so I at least explain the notation.

Iter 1:

$c_1 = (2, 1)$, $c_2 = (8, 5)$:

* $x_1 \rightarrow c_1 = 1$, $x_2 \rightarrow c_1 = 0$, $x_3 \rightarrow c_1 = 8$, $x_4 \rightarrow c_1 = 32$, $x_5 \rightarrow c_1 = 50$, $x_6 \rightarrow c_1 = 52$
 * $x_1 \rightarrow c_2 = 65$, $x_2 \rightarrow c_2 = 52$, $x_3 \rightarrow c_2 = 20$, $x_4 \rightarrow c_2 = 4$, $x_5 \rightarrow c_2 = 2$, $x_6 \rightarrow c_2 = 0$

Assignments are: $[1, 1, 1, 2, 2, 2]$

New Centroids:

$$c_1^* = ((1+2+4) / 3, (1+1+3) / 3) = (7/3, 5/3)$$

$$c_2^* = ((6+7+8) / 3, (5+6+5) / 3) = (7, 16/3)$$

Iter 2:

$c_1^* = (7/3, 5/3)$, $c_2^* = (7, 16/3)$:

* $x_1 \rightarrow c_1 = 20/9$, $x_2 \rightarrow c_1 = 5/9$, $x_3 \rightarrow c_1 = 41/9$, $x_4 \rightarrow c_1 = 221/9$,
 $x_5 \rightarrow c_1 = 365/9$, $x_6 \rightarrow c_1 = 389/9$
 * $x_1 \rightarrow c_2 = 493/9$, $x_2 \rightarrow c_2 = 394/9$, $x_3 \rightarrow c_2 = 130/9$, $x_4 \rightarrow c_2 = 10/9$,
 $x_5 \rightarrow c_2 = 4/9$, $x_6 \rightarrow c_2 = 10/9$

Assignments are: $[1, 1, 1, 2, 2, 2]$

The assignments did change so this means k-means has converged.

b) Number of iterations = 2

Problem 5 12 pts: Consider the following dataset in $\mathbb{R}^2$:

Point	x_i [1]	x_i [2]
x_1	1	1
x_2	2	1
x_3	5	4
x_4	6	5

Perform agglomerative hierarchical clustering using Euclidean distance and complete linkage. Stop when all points are in one cluster. Show your work and describe the merging sequence.

Answer:

In order to form the clusters we first calculate all the euclidean distances between each pair of points. The notation convention below is d_{mn} where m and n represent the 1 of the 4 points above.

$$d_{12} = ((2-1)^2 + (1-1)^2)^{0.5} = 1$$

$$d_{13} = ((5-1)^2 + (4-1)^2)^{0.5} = 5$$

$$d_{14} = ((6-1)^2 + (5-1)^2)^{0.5} = 41^{0.5}$$

$$d_{23} = ((5-2)^2 + (4-1)^2)^{0.5} = 18^{0.5}$$

$$d_{24} = ((6-2)^2 + (5-1)^2)^{0.5} = 32^{0.5}$$

$$d_{34} = ((6-5)^2 + (5-4)^2)^{0.5} = 2^{0.5}$$

Using the distances above our merge sequence is,

1. Merge x_1 and x_2 because those two points are the closest, $d_{12} = 1$. Thus let $C_{12} = \{x_1, x_2\}$

2. Given the cluster C_{12} , x_3 , and x_4 we merge x_3 and x_4 because

$$D(C_{12}, x_3) = \max(d_{13}, d_{23}) = 5$$

$$D(C_{12}, x_4) = \max(d_{14}, d_{24}) = 41^{0.5}$$

$$D(x_3, x_4) = d_{34} = 2^{0.5}$$

d_{34} has the smallest distance above. Our second cluster is now $C_{34} = \{x_3, x_4\}$

3. There are only two clusters so we merge C_{12} and C_{34} to have one cluster.

Problem 6 **6 pts**: In your own words, describe at least two key differences between clustering and classification.

Answer:

- * The goal of the approaches with respect to the dataset of a problem differ. Classification aims to predict a label given the data while clustering reveals what groups exist in a dataset via mathematical metrics and algorithms without assuming what the groups are.
- * The labels predicted in classification already have a definition / understanding in the context of the problem. On the other hand, after clustering, one will interpret the meaning of the clusters.

Problem 7 (10 pts): In Figure 2 is the result of applying a clustering algorithm on a dataset where each datapoint has two features. Each cluster has a center marked by a black $\times$.

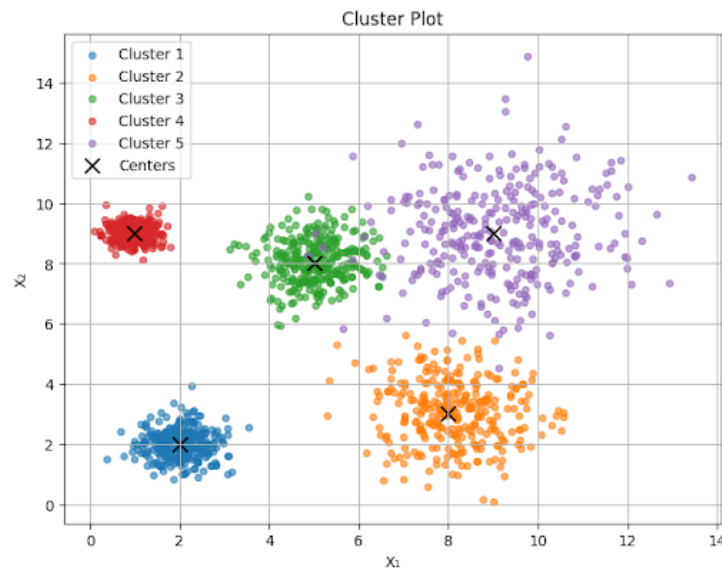


Figure 2: Clustering Data

Not all of the clustering methods that we have covered could create these clusters. Which of the following clustering methods **could** create these clusters? Justify your answer(s).

- (a) K-means
- (b) DBSCAN
- (c) GMM
- (d) Agglomerative Clustering

Answer:

- a) No. There are purple points closer to the green center that would have been assigned to the green cluster.
- b) No. DBSCAN has a hard time with cluster of different densities given it uses a single delta to determine membership in a cluster.
- c) Yes. Each cluster can have different variance, which means the points can spread out like the purple ones. The points aren't guaranteed to be assigned to the closest center based on a distance metric.
- d) No. The algorithm doesn't produce overlapping shapes that would happen as a result of those purple points mixed in the cluster of green points.

Problem 8 12 pts: You are given a dataset of four 2D data points:

Point	$x_i [1]$	$x_i [2]$
x_1	2	0
x_2	0	2
x_3	3	1
x_4	1	3

Perform Principal Component Analysis (PCA) on this dataset by hand to reduce it to one principal component. Show all steps clearly.

Answer:

1. First center the data, $\mu = (1.5, 1.5)$

The centered data is $Z = \begin{bmatrix} .5 & -1.5 \\ -1.5 & .5 \\ 1.5 & -.5 \\ -.5 & 1.5 \end{bmatrix}$

2. The cov. matrix is

$$\Sigma = \frac{1}{4} Z^T Z = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$$

3. Find eigenvalue & vectors. There's easier solving $\det(\Sigma - \lambda I) = 0 \Rightarrow \lambda^2 - 5/2 \lambda + 1 = 0 \Rightarrow \lambda_1 = 2, \lambda_2 = 1/2$

Using $\lambda_1 = 2$ we easily find the eigen vector $(1, -1)$. After normalizing

we get $1/\sqrt{2} (1, -1)$, which is the 1st comp.

Note: there was so little space above.

Problem 9 9 pts: A company manufactures electronic components and wants to monitor defective parts on its assembly line. Each part is measured for length, width, and weight. The quality control team decides to use a Gaussian Mixture Model (GMM) to cluster the parts into acceptable and defective clusters. They follow these steps:

- (a) Create a dataset consisting of the part features: length, width, and weight. Because they want to detect differences in the features, they decide not to standardize the data so that the defective differences are obvious to the GMM.
- (b) Because they want to detect defective parts, they decide to use exactly 2 components, creating two clusters: 'acceptable' and 'defective.'
- (c) Because they know GMMs are sensitive to initialization, they perform multiple restarts.

For each of these steps, state whether it is correct or incorrect, and explain why.

Answer:

a) Incorrect. If the one of the features has a much larger scale (e.g. weight) relative to the others and the defects are often more correlated with other features then the feature with the larger scale can end up dominating.

b) Incorrect. The statement makes the incorrect assumption that the acceptable and defective parts fall into just two clusters detected by GMM. In fact there could be outliers in the subgroups by the features that don't get correctly classified under a 2-component model.

c) Correct. GMM can get stuck in local optima.

Problem 10 **10 pts**: In 1–2 paragraphs, discuss a problem that you’re interested in solving (this could be related to a work problem, a potential personal project, or just something that interests you) that could be solved with clustering or visualized/simplified with PCA. What kind of data is involved? If solving using clustering, what algorithms would you try and why? If using PCA, how will the new principal components help you solve your problem?

Answer:

Problem: Identify anomalous system access.

Description: Explore groups and features that would help identify anomalous system access.

Data per user: date, ip, #logins, #failed logins, #distinct login locations, #devices, #previous lockouts, #hours since last login, #days since password reset, # distinct locations, .. lots more

Model: Use PCA to identify the components that explain normal access. Use the high reconstruction error as an indicator of suspicious access. Additionally I'd try,

- * DBSCAN would be valuable to try because noise points are the anomalies.
- * GMM + PCA because some of the data would be heavy-tailed, which GMM isn't made for. (e.g. # logins per user)

Problem 11 (9 pts): A bank uses a machine learning model to predict whether loan applicants will repay their loans. The model outputs 1 if it predicts the applicant will repay the loan and 0 if it predicts they will default (a classification problem). The bank wants to ensure that the model is fair across two demographic groups: Group A and Group B. According to the Equal Opportunity definition of fairness, a model is fair if the True Positive Rate (TPR) for each group is within a threshold ϵ of each other. For this problem, let $\epsilon = 0.05$. You are given the following confusion matrices for each group based on a test dataset:

Group A

	Predicted: Repay (1)	Predicted: Default (0)
Actual: Repay (1)	80	20
Actual: Default (0)	10	90

Group B

	Predicted: Repay (1)	Predicted: Default (0)
Actual: Repay (1)	45	55
Actual: Default (0)	5	95

- Compute the True Positive Rate (TPR) for Group A and Group B and determine whether the model satisfies the Equal Opportunity fairness criterion. Show your work.
- Propose a simple method the bank could use to adjust the classification for two groups to achieve Equal Opportunity in practice, assuming they start with a model that does not satisfy the criterion.
- Discuss any trade-offs or potential risks of adjusting classifications separately for each demographic group.

Answer:

- a) Group A TPR = $80 / (80 + 20) = 0.8$
 Group B TPR = $45 / (45 + 55) = 0.45$

According to equal opportunity fairness act the TPR's should be within 0.05 however the difference = $0.8 - 0.45 = 0.35 > 0.05$. Thus criterion isn't fulfilled.

- b) A simple method the bank could use to adjust classification for the two groups is to use different decision thresholds. Doing that would affect the TPR.
- c) The trade-offs of adjustments would be,
- * Lowering thresholds could lead to more defaults
 - * Group specific thresholds can lead to unfairness to individuals as we identified in previous readings.
 - * Typically attempts to be fair require the use of sensitive data like age, race, gender etc that may have legal restrictions.

Problem 12 **3 pts**: How many hours do you estimate you spent on this assignment? (graded on completion of (a), (b), and (c))

(a) Coding:

(b) Written:

(c) Reading:

Answer:

Coding: ~4hrs

Written: ~7hrs

Reading: ~10hrs

Problem 13 **Gratitude pts**: Any feedback you would like to share (optional)?

Answer:

Working with the PDF's or latex is very time consuming. Being able to do the written work in a Jupyter Notebook is preferable.